

Evidence-Aware Research Communication with Retrieval

SARAL Prototype Demonstration | September 2026

Abstract

Researchers often need to explain a paper to audiences with different levels of technical knowledge. This demonstration paper describes a retrieval-backed communication workflow. The workflow maps generated statements to source chunks so readers can inspect the evidence behind each statement.

1. Problem

Summaries are useful only when a reader can tell which evidence supports them. A policymaker may need a concise explanation of public value, while graduate students may need methods, assumptions, and equations. A single fixed summary does not serve both audiences well.

2. Method

The system splits a paper into page-aware chunks and ranks the chunks against an audience-specific request. A generator uses only selected evidence to create slide bullets, speaker notes, and a script. Each generated item stores one or more chunk identifiers and page numbers.

CitationCoverage = (generated claims with at least one citation) / (total generated claims)

We use lexical grounding as a lightweight factuality proxy. For a generated claim and its retrieved evidence, the score is the fraction of unique claim tokens that also occur in the evidence.

Grounding(claim, evidence) = shared unique tokens / unique claim tokens

3. Illustrative Findings

In a small internal demonstration, evidence-linked outputs made it easier to verify statements during review. Graduate-student outputs retained the evaluation definitions, while policymaker outputs foregrounded transparency and accountability. Overlap metrics are not proof of factuality; human review remains necessary for high-stakes use.

4. Audience-Specific Requests

Audience	Primary need	Recommended output
Policymakers	Public value and traceability	90-second plain-English script
Graduate students	Methods and equations	Technical five-slide talk
Press	Clear, quotable framing	30-second press summary

5. Limitations

Sparse retrieval can miss relevant paraphrases, and a generated citation does not itself guarantee that a claim is correct. A production system should add semantic retrieval, independent citation checking, safety review, feedback collection, and access controls.

6. Conclusion

Retrieval-backed generation can make research communication more adaptable while preserving a clear path back to the underlying paper. The strongest demonstration is an iterative edit that changes wording for an audience without losing evidence links.